

General Aspects of Concurrent Aerobic and Strength Training for Performance in Team Sports

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Introduction

Many team sports require players to perform high-intensity efforts repeatedly and intermittently in competition. Specific training strategies such as concurrent strength and aerobic training are proposed to improve players' ability to repeat high-intensity efforts [1]. Development of the aerobic system aids in improving recovery between efforts by replenishing phosphocreatine and improving muscle buffer capacity, factors which are critical to performance in team sports [2, 3]. The aerobic energy system is also suggested to aid in post-match recovery following match-play. This quality has implications for team sport schedules where multiple matches are performed over a week (e.g. basket-ball, handball, volleyball, rugby, futsal, water polo) or a day and over consecutive days with only few hours separating matches (rugby sevens) [1]. While there is good evidence to suggest that resistance training could be beneficial for performance in a single maximal effort (sprint, jump, tackle etc.), the impact of this training format on player capacity to successively repeat these efforts is less clear [3]. However, several studies have reported that resistance training produces similar increases in mean work capacity during a repeated sprint test (~12%) [4] compared with high-intensity interval training (~13%) [5]. Resistance training also improved both initial sprint performance (8–9%) and the sprint decrement score (~20%) [5] in a repeated sprint test. The increases in repeated sprint ability (RSA) previously reported are likely to be linked to strength gains [3].

The issue of concurrent strength and aerobic training is seemingly at the center of attention of many contemporary team sport practitioners. A survey led by Jones et al. [6] examined training and monitoring strategies of strength and conditioning coaches of rugby union players. The results showed that 77% of respondents accounted for the aforementioned interference effect while programming training

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for rugby union players. In addition, 47% of coaches believed it was very important to consider possible incompatibilities between aerobic and strength training when generating conditioning plans. The findings of this survey emphasized periodization would be the most effective means of avoiding any potential effect of aerobic type stimulus on strength and power development [6].

Influence of Concurrent Training Variables on the Interference Effect: A Case Study with the French Rugby Union

Previous studies showed that strength training combined with aerobic exercises in a single program is known to impair strength and power gains in comparison with strength training alone, if training frequency is high [7–9]. This was discussed in previous chapters of the present book as well as in a meta-analysis by Wilson et al. [10]. Modality, duration and frequency of endurance exercises were identified as the main factors supporting the interference effects of endurance training on the expected improvement of strength and power in response to concurrent resistance training. Moreover, the influence of several training factors such as intensity and volume of concurrent exercises as well as the order of training sessions have been previously addressed to minimize this neuromuscular interference [11–13]. An important factor potentially affecting the magnitude of interference in team sport athletes might also be the duration of the recovery period between strength and endurance exercises.

Recovery Delay Between Sequences

In a previous study in rugby union players [14], we assessed whether the recovery between strength and aerobic high-intensity interval exercises would determine the magnitude of possible interference. The sequence of performing strength training prior to aerobic exercise was chosen to avoid residual fatigue induced by the metabolic session and, therefore, maintaining the intensity of strength training. The question of recovery duration is warranted because the training load in elite team sport has increased over recent years. Sessions are often scheduled twice-a-day, separated by none or only a few hours. In our aforementioned study, daily training without recovery between sequences and, to a lesser extent, training twice-a-day with 6 h recovery between strength and aerobic training sessions did not seem to be optimal for strength and power gains, nor for adaptations in $\text{VO}_{2\text{peak}}$, while daily training with 24-h recovery between sessions appeared to be favorable. These results suggest that strength and conditioning coaches should avoid scheduling two different exercise modalities (strength vs. aerobic) in close proximity but rather aim for a 24-h recovery period in order to assure optimal neuromuscular and cardiovascular adaptations.

In team sports like rugby, coaches also have to program specific technical and tactical training sessions. These can generate high aerobic demands similar to those observed during traditional endurance training. Thus, care should be taken when scheduling specific training sessions in close proximity to strength exercise sessions. Therefore, it is recommended to monitor the specific technical and tactical training load and the induced physiological responses, e.g., by means of global

positioning system technology, and portable HR monitors. These devices allow regulation of the intensity of specific training and, thus, may help to minimize the risk of interference in strength development [14].

Type of Aerobic Training

High-intensity interval training (HIT) of various modes (i.e., different lengths of exercise and recovery bouts) is today considered one of the most effective means of improving cardiorespiratory and metabolic function and, in turn, the physical performance of athletes. HIT involves repeated short-to-long bouts of high-intensity exercise interspersed with recovery periods (short and long intervals). For team sport athletes, the inclusion of sprints and all-out efforts into HIT programs has also been shown to be an effective methodology. These types of HIT (short and long intervals, repeated sprint and sprint interval training) elicit different acute physiological metabolic responses and neuromuscular strain [15]. In light of this, it would seem relevant to determine whether the type of endurance exercise would influence the magnitude of interference on neuromuscular adaptations. The research and development department of the French Rugby Union Federation investigated the impact of 2 HIT protocols (i.e., short interval vs. sprint interval, Fig. 26.1) in amateur rugby sevens players, performed over 8 weeks on maximal strength and power, and RSA [16] (Fig. 26.1). The main finding of this study was that strength development was not compromised following concurrent training with short intervals (INT) but showed a slight impairment of slow concentric torque production gains following concurrent training with sprint interval (SIT). However, while strength gains were somewhat compromised in the strength and SIT group, this training actually led to the largest gains in VO₂ peak. Thus, according to these findings, coaches may need to carefully consider the needs of a given athlete as both optimal VO₂ peak and strength gains may be difficult to achieve in amateur rugby sevens players. Tables 26.1 and 26.2 gives an example of how rugby

		Monday	Tuesday	Wednesday	Thursday	Friday
Familiarization		Session 1			Session 2	
Initial tests		MVC		Graded maximal aerobic test		RSA
Training duration 8 weeks	SIT	Strength	Sprint-interval		Strength	Sprint-interval
	INT	Strength	Short-interval		Strength	Short-interval
	CON	Strength	–		Strength	–
Final tests		MVC		Graded maximal aerobic test		RSA

Fig. 26.1 Experimental design of the study. Abbreviations: *CON* strength training, *INT* concurrent strength and short interval training, *MVC* maximal voluntary contraction, *RSA* repeated sprint ability, *SIT* concurrent strength and sprint interval training. Sprint interval workload: 4–8 run all-out efforts interspaced by 4 min passive recovery. Short interval workload: two series of 8–12 min 30 s run at maximal aerobic velocity (MAV) interspaced by 30 s of active recovery at 50% MAV, 3 min of passive recovery between the series

Table 26.1 A sample rugby sevens weekly training schedule with short-/long interval training

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	Well-being/ Medical Resistance (LB)	Warm-up Rugby—tech/tact Aerobic (short/long interval)	Warm-up Agility/Sprint Rugby—tactical Recovery	Warm-up Rugby—tech/tact Aerobic (short/long interval)	Warm-up Core training Running skills	Warm-up Rugby—tech Active recovery (swimming pool)	
Afternoon	Nap Rugby—tactical Recovery	Nap Resistance (UB) Recovery		Nap Resistance (L&UB) Recovery	Nap Rugby—tactical Aerobic suppl. Recovery		

Table 26.2 A sample rugby sevens weekly training schedule with sprint interval training (SIT)

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	Well-being/ Medical Resistance (L&UB)	Warm-up Rugby—tactical	Warm-up Running skills Rugby—technical Recovery	Warm-up Resistance (L&UB)	Warm-up Rugby—tech/tact	Warm-up Rugby—tech Active recovery (swimming pool)	
Afternoon	Nap Running skills Rugby—tactical Recovery	Nap Core training Aerobic (SIT) Recovery		Nap Rugby—tactical Recovery	Nap Core training Aerobic (SIT) Recovery		

sevens schedule a weekly training INT or SIT. The observed interference effect could be explained by an exacerbated residual fatigue mechanism. While we allowed 48 h of rest between subsequent training sessions in this study, Howatson and Milak [17] suggested that this may be insufficient. In their study, it was shown that repeated sprint exercises may induce a significant decrement of quadriceps MVC and a significant increase in muscle soreness lasting for up to 48 h. These results further confirm that sprint interval training-induced muscle fatigue can impair the quality of a subsequent resistance training sessions, even though this is performed multiple days later.

Therefore, it is recommended to consistently monitor the fatigue generated by repeated sprint or sprint interval training in order to limit the likelihood of neuromuscular interference. Many variables, such as increased recovery time between sequences or different recovery strategies (ice bath, cryotherapy, and nutritional strategies) can be manipulated to optimize recovery at baseline performance. The one-sequence a day training configuration, with prolonged recovery between aerobic and strength exercises has to be programmed especially when HIT is composed of repeated short or long sprints. Another solution is to reduce muscle soreness and damage and, therefore, exacerbated residual fatigue caused through intense efforts by performing SIT on a bicycle. Indeed, it was shown that performing concurrent cycle sprint interval and resistance training does not seem to attenuate the strength response when compared to resistance training alone [18].

Effects of Concurrent Training During Pre- and In-Season in Other Team Sports

The effects of pre- and in-season concurrent training on maximal strength and power have been observed in other team sports during real training conditions and account for technical and tactical workload. The effects of two different HIT programs (long intervals, i.e., 4×4 min vs. short intervals, i.e., 16×100 m) performed concurrently with maximum strength and specific water polo training during pre-season has been compared [19]. The results demonstrated that improvements in swimming endurance parameters and maximum strength occurred following both programs in national elite players. It should, however, be noted that the training frequency with two combined sessions per week was rather low but may have been adequate and effective to elicit significant strength and endurance increments in national elite water polo players [19]. Similar results were also shown in futsal players performing concurrent strength and repeated sprint training (RST). The main results of this study emphasized faster mean RSA time following concurrent training than after only specific futsal training [20]. Hence, additional strength and RST can be an effective strategy to improve futsal-specific performance.

Although strength and power may be relatively easy to develop during preseason training periods as observed in the two previous studies [21, 22], there is some disagreement as to whether preseason levels of strength and power can be maintained during the long in-season playing periods, especially when a large amount of energy system (aerobic and anaerobic) conditioning or lengthy team practices are

performed [23–25]. Scheidner et al. [23] and dos Remedios et al. [24] reported a significant reduction in maximal strength 13–14 weeks into the in-season in college football players. Legg and Burnham [25] reported losses in shoulder strength by as much as 25% over the course of a 10-week in-season period in college-aged football players. Beyond aiming to develop neuromuscular qualities in-season, the main objective of concurrent training for team sport athletes should, therefore, be to maintain strength, power, and speed performances. In this way, Baker [26] investigated the effects of an in-season program of concurrent training in professional rugby league football players and noted that maximal strength and power can be maintained at the maximum preseason levels for long periods of up to 29 weeks. The key to maintaining strength and power during the in-season may lie in:

- Having athletes initially better conditioned to perform concurrent training
- It has been postulated that athletes who typically perform little metabolic aerobic training in the preseason and who may possess low aerobic capacities may experience greater decreases in maximal strength during the in-season compared to athletes with a long training history of concurrent strength and aerobic training [26]. If athletes are better conditioned to perform concurrent resistance and endurance training, then the interference effects of training, game, and practice demands on strength development or maintenance may be reduced to some degree.
- A sufficient workload
- According to practical recommendations in professional soccer players edited by Rønnestad et al. [27], one strength maintenance session per week, during a 12-week period, may be sufficient to maintain initial gain in strength and sprint performance achieved during a preceding preparatory period. In contrast, performing only 1 strength maintenance session every 2 weeks seemed too low to maintain leg strength and 40-m sprint performance.
- The prioritization of training goals, sequencing, and timing of training
- Leveritt and Abernethy [28] suggested that training goals need to be prioritized so that the primary training goal should be trained first in an unfatigued state. This prioritization of training goals would then dictate the sequencing and/or timing of other training sessions.
- Utilizing an appropriate periodization model that allows for periods of high, medium, and low training volumes and intensities. This variation of training load would avoid overreaching and stimulate physiological adaptations [22].

A common training method in team sport, called small-sided games (SSGs), has recently become the focus of scientific research because of its ability to develop physical capacities together with sport-specific tactical and technical skills [29]. The primary benefits of SSGs training are that the game can replicate the movement patterns, physiological demands, and technical requirements of competitive match play [29], while also requiring players to make decisions under pressure and fatigue. In addition, compared with traditional fitness training sessions, SSGs training is thought to increase player compliance and motivation because it is perceived to be sport-specific exercise that maximizes the training time spent with the ball [29].

Iacono et al. [30] have shown that HIT and SSGs training were equally effective in developing aerobic capabilities and supra-maximal intermittent performance (according to the Yo-Yo intermittent recovery level 1 test) in elite handball players. Moreover, 8 weeks of SSGs training led to greater improvement in anaerobic performance assessed by 10 and 20-m sprints, vertical jumps and agility test compared with HIT. Consequently, during the in-season period, handball coaches may prefer the SSGs training instead of HIT to optimize anaerobic and aerobic training. Finally, coaches and players could simultaneously stimulate higher physical demands than those presented in official match-play (maximal aerobic power, changes of directions, and high-intensity actions) to continually develop handball players' fitness components while also encompassing handball-specific technical and tactical elements. Other studies, achieved notably in junior elite basket [31] and volleyball players [32], corroborated these results emphasizing significant improvements in fitness and technical skills following the SSGs training period. Therefore, SSGs could be considered as an optimal training method during in-season period, not only to maintain but also develop aerobic and anaerobic physical qualities.

Summary

In conclusion, concurrent training variables such as the exercise order, recovery period as well as the type and modality (run or cycle) of HIT may influence the interference effect. Practitioners have to keep in mind that the training program should be structured according to the prioritized objective. Moreover, they have to be cautious to aid the recovery process when recovery time is restricted.

Strategies and objectives with regard to managing concurrent training-induced interference should be different according to the phase of the season. In preseason, the main objective is to maximize the development of physical qualities whereas practitioners seek to maintain these during in-season period. However, we emphasized that SSGs training could allow development of both aerobic and anaerobic qualities (through agility and vertical jump) even in the competitive phase. Its effect on muscular strength however remains to be proven.

All previous studies mentioned in this chapter showed the large complexity when implementing concurrent training. Sometimes, this would be mainly dependent on the coach's preferences, the philosophy of the sports science staff, the demands of the competition, and the availability of facilities [33]. The aim of further studies should be to stimulate the interest of practitioners to objectively develop periodization models for concurrent aerobic and strength training, especially for team sport athletes.

References

1. Ross A, Gill N, Cronin J. Match analysis and player characteristics in rugby sevens. *Sports Med.* 2014;44:357–67.
2. Tomlin DL, Wenger HA. The relationship between aerobic fitness and recovery from high intensity intermittent exercise. *Sports Med.* 2001;31:1–11.

3. Bishop D, Girard O, Mendez-Villanueva A. Repeated-sprint ability part II: recommendations for training. *Sports Med.* 2011;41:741–56.
4. Edge J, Bishop D, Goodman C. Effects of high- and moderate-intensity training on metabolism and repeated sprints. *Med Sci Sports Exerc.* 2005;37:1975–82.
5. Edge J, Hill-Haas S, Goodman C, et al. Effects of resistance training on H⁺ regulation, buffer capacity, and repeated sprints. *Med Sci Sports Exerc.* 2006;38:2004–11.
6. Jones TW, Smith A, Macnaughton LS, French DN. Strength and conditioning and concurrent training practices in elite rugby union. *J Strength Cond Res.* 2016;30:3354–66.
7. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. *Eur J Appl Physiol.* 1980;45:255–63.
8. Dudley GA, Djamil R. Incompatibility of endurance- and strength-training modes of exercise. *J Appl Physiol.* 1985;59:1446–51.
9. Hennessy LC, Watson AWS. The interference effects of training for strength and endurance simultaneously. *J Strength Cond Res.* 1994;8:12–9.
10. Wilson JM, Marin PJ, Rhea MR, Wilson SMC, Loenneke JP, Anderson JC. Concurrent training: a meta-analysis examining interference of aerobic and resistance exercises. *J Strength Cond Res.* 2012;26:2293–307.
11. Collins MA, Snow TK. Are adaptations to combined endurance and strength training affected by the sequence of training? *J Sports Sci.* 1993;11:485–91.
12. Chtara M, Chamari K, Chaouachi M, Chaouachi A, Koubaa D, Feki Y, Millet GP, Amri M. Effects of intra-session concurrent endurance and strength training sequence on aerobic performance and capacity. *Br J Sports Med.* 2005;39:555–60.
13. Chtara M, Chaouachi A, Levin GT, Chaouachi M, Chamari K, Amri M, Laursen PB. Effect of concurrent endurance and circuit resistance training sequence on muscular strength and power development. *J Strength Cond Res.* 2008;22:1037–45.
14. Robineau J, Babault N, Piscione J, Lacomme M, Bigard AX. The specific training effects of concurrent aerobic and strength exercises depend on recovery duration. *J Strength Cond Res.* 2016;30:672–83.
15. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: part I: cardiopulmonary emphasis. *Sports Med.* 2013;43:313–38.
16. Robineau J, Lacomme M, Piscione J, Bigard X, Babault N. Concurrent training in Rugby Sevens: effects of high-intensity interval exercises. *Int J Sports Physiol Perform.* 2017;12:336–44.
17. Howatson G, Milak A. Exercise-induced muscle damage following a bout of sport specific repeated sprint. *J Strength Cond Res.* 2009;23:2419–24.
18. Cantrell GS, Schilling BK, Paquette MR, Murlasits Z. Maximal strength, power, and aerobic endurance adaptations to concurrent strength and sprint interval training. *Eur J Appl Physiol.* 2014;114:763–71.
19. Botonis PG, Toubekis AG, Platanou TI. Concurrent strength and interval endurance training in Elite Water Polo Players. *J Strength Cond Res.* 2016;30:126–33.
20. Phongsri K, Tongnillpant N, Ajjimaporn A, Silalertdetkul. Effect of additional concurrent specific strength and repeated sprint training during preseason on performance in futsal players. In: ECSS MetropolisRuhr; 2017.
21. Baker D. Applying the in-season periodisation of strength and power training to football. *Strength Cond J.* 1998;20:18–24.
22. Fleck SJ, Kraemer WJ. Designing resistance training programs. Champaign: Human Kinetics; 1997.
23. Scheidner V, Arnold B, Martin K, Bell D, Crocker P. Detraining effects in college football players during the competitive season. *J Strength Cond Res.* 1998;12:42–5.
24. Dos Remedios KA, Dos Remedios RL, Loy SF, Holland GF, Vincent WJ, Conley LM, Hing M. Physiological and field test performance changes of community college football players over a season. *J Strength Cond Res.* 1995;9:211–5.
25. Legg D, Burnham D. In-season shoulder abduction strength changes in football players. *J Strength Cond Res.* 1999;13:4381–3.

26. Baker D. The effects of an in-season of concurrent training on the maintenance of maximal strength and power in professional and college-aged rugby league football players. *J Strength Cond Res.* 2001;15:172–7.
27. Rønnestad BR, Nymark BS, Raastad T. Effects of in-season strength maintenance training frequency in professional soccer players. *J Strength Cond Res.* 2011;25:2653–60.
28. Leveritt M, Abernethy PJ. Acute effects of high-intensity endurance exercise on subsequent resistance activity. *J Strength Cond Res.* 1999;13:47–51.
29. Buchheit M, Laursen PB, Kuhnle J, Ruch D, Renaud C, Ahmaidi S. Game-based training in young elite handball players. *Int J Sports Med.* 2009;30:251–8.
30. Iacono AD, Eliakim A, Meckel Y. Improving fitness of elite handball players: small-sided games vs. high-intensity intermittent training. *J Strength Cond Res.* 2015;29(3):835–43.
31. Delextrat A, Martinez A. Small-sided game training improves aerobic capacity and technical skills in basketball players. *Int J Sports Med.* 2014;35(5):385–91.
32. Gabbett TJ. Do skill-based conditioning games offer a specific training stimulus for junior elite volleyball players? *J Strength Cond Res.* 2008;22(2):509–17.
33. Enright K, Morton J, Iga J, Drust B. The effect of concurrent training organisation in youth elite soccer players. *Eur J Appl Physiol.* 2015;115:2367–81.